Assumptions and the evidence behind them

Every assumption the LILA BLACK Journey Viewer makes about the data, with the evidence for it underneath. Everything here is taken from the files in `docs/evidence/`; no number was added that isn't in one of them. Each section links to its source.

Status labels

- **Current**: what the pipeline and site do today.
- **Superseded**: the evidence still stands, but the decision built on it changed later (the newer rule is given).
- **Contested**: the forensic audit found that the evidence doesn't fully support it. It is still in use, and the decision is S's to make.

Read this first: some evidence files are older than the current rules.

- `event_forensics/*.md` and the audit say the 17 numeric-ID journeys are `unknown`. Since 2026-10-24 they are counted as **human** and flagged `id_mismatch` (the CSV, `forensics_summary.json` and `build_report.json` were regenerated; the `.md` pages were not).
- `spatial_analysis/`, `zone_sensitivity.md`, `interpretation_layer/` and `tabs_and_outlines/` show the earlier hotspot rule ("consistent concentrations"). It was replaced on 2026-10-24 (see A17).
- Screenshots in `heatmap_review/`, `interpretation_layer/` and `tabs_and_outlines/` show earlier versions of the UI.

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Appendix: open points from the audit · index of every evidence file

A1. `ts` holds Unix seconds, not milliseconds

Status: Current. **Assumption:** `ts` is stored as `timestamp[ms]` but holds Unix epoch **seconds**. Match time = `ts` − the earliest `ts` in the match.

Evidence

- Read as milliseconds (the declared type), the whole dataset spans **6 min 40 s on 1970-01-21**, and a 65-row file would last under 1 s.
- Read as seconds, rows fall on **2026-02-09 23:58 to 2026-02-14**, matching the five folder names. Each file spans **13–890 s**, about a 15-minute game.
- Movement rows come about every 5 s: **42,816** human intervals are exactly 5 s.
- The data README says `ts` is milliseconds elapsed within the match. The audit marks this as **contradicted**.
- Audit verdict: *Trust (I, very strong)*.

Sources: INDEPENDENT_EVIDENCE.md §1.6, §2.1, §6.1 · RECONCILIATION_EVIDENCE.md §1

A2. The `event` column is UTF-8 text

Status: Current.

Evidence

- The bytes are pure ASCII and decode strictly as UTF-8.
- There are 8 values: Position 51,347; BotPosition 21,712; Loot 12,885; BotKill 2,415; BotKilled 700; KilledByStorm 39; Kill 3; Killed 3.

Source: INDEPENDENT_EVIDENCE.md §1.3

A3. The README coordinate formula is correct, with the y-flip

Status: Current (decision D4). **Assumption:** `u = (x − origin_x)/scale`, `v = (z − origin_z)/scale`, `px = u·W`, `py = (1 − v)·H`, using each minimap's **real** size (AmbroseValley 4320², GrandRift 2160 × 2158, Lockdown 9000²; not 1024 px as the README says). `y` is height and is ignored.

Evidence: maths (the pass/fail check)

- The README's worked example gives (77.995, 890.368), which is the expected (78, 890).
- World corners land on image corners. GrandRift's world centre lands at (1080, 1079) on its non-square image.
- +x moves right and +z moves up.
- `tests/test_projection.py` runs these checks for all three maps.

Evidence: content registration (independent audit) Every movement sample was mapped and checked for whether it lands on map content.

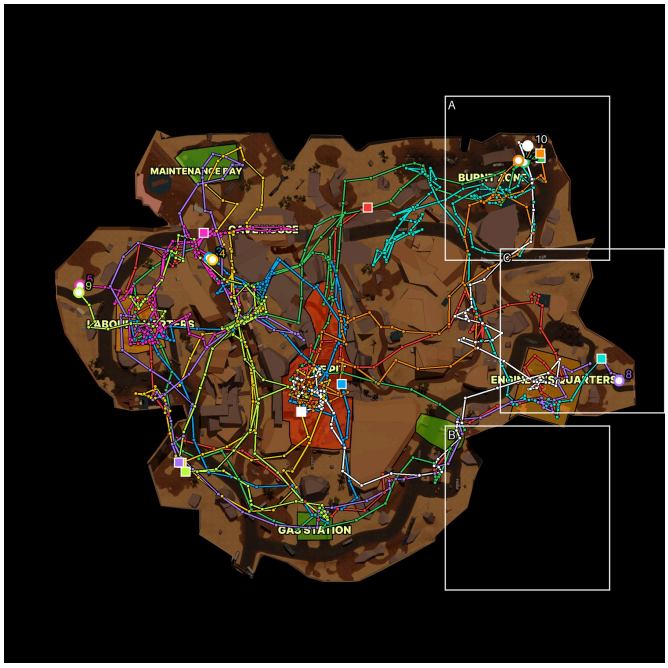
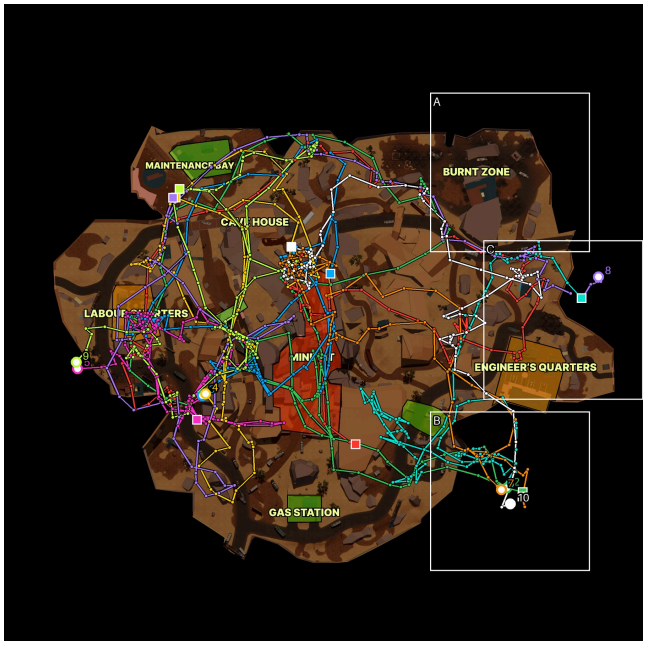
Map	On content (README parameters)	Best alternative flip
AmbroseValley	100.00 %	95.7 % (no z-flip)
GrandRift	99.69 %	95.7 % (x mirrored)
Lockdown	99.99 %	97.4 % (x mirrored)

- Scanning scale ±10 % and shifts ±30 units finds no better fit.
- Audit verdict: trust **to about ±10 world units**. Zero rotation is assumed, not tested.

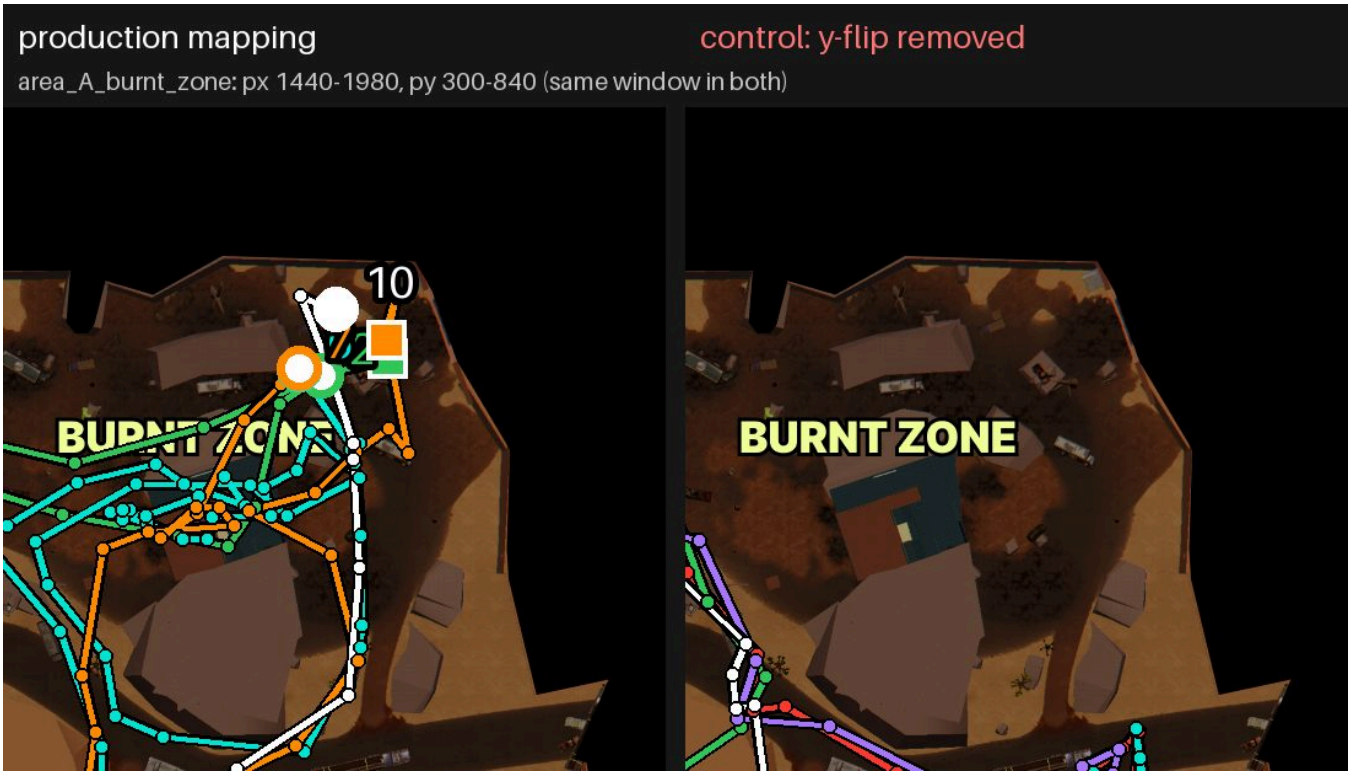
Evidence: pictures (supporting, not pass/fail)

- With the production mapping, ten long GrandRift journeys (1,046 positions) start and end inside the map outline and cluster on named places: Labour Quarters, Mine Pit, Engineer's Quarters, Gas Station, Burnt Zone.
- With the y-flip removed, start markers fall into the black void.
- Near-black share (descriptive only): 4.2 % production against 11.5 % no-flip.

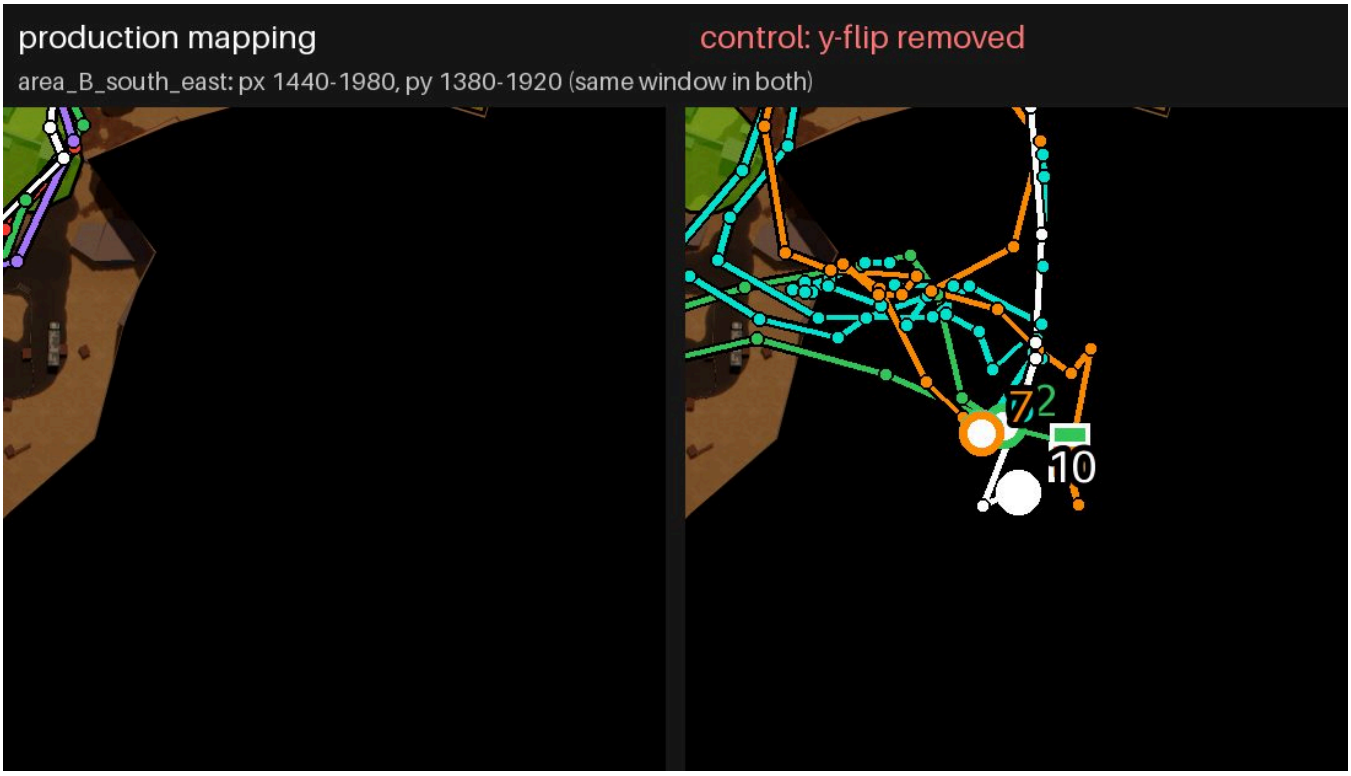
Full GrandRift map: production (left) vs y-flip removed (right)

Production	Control: no y-flip
	

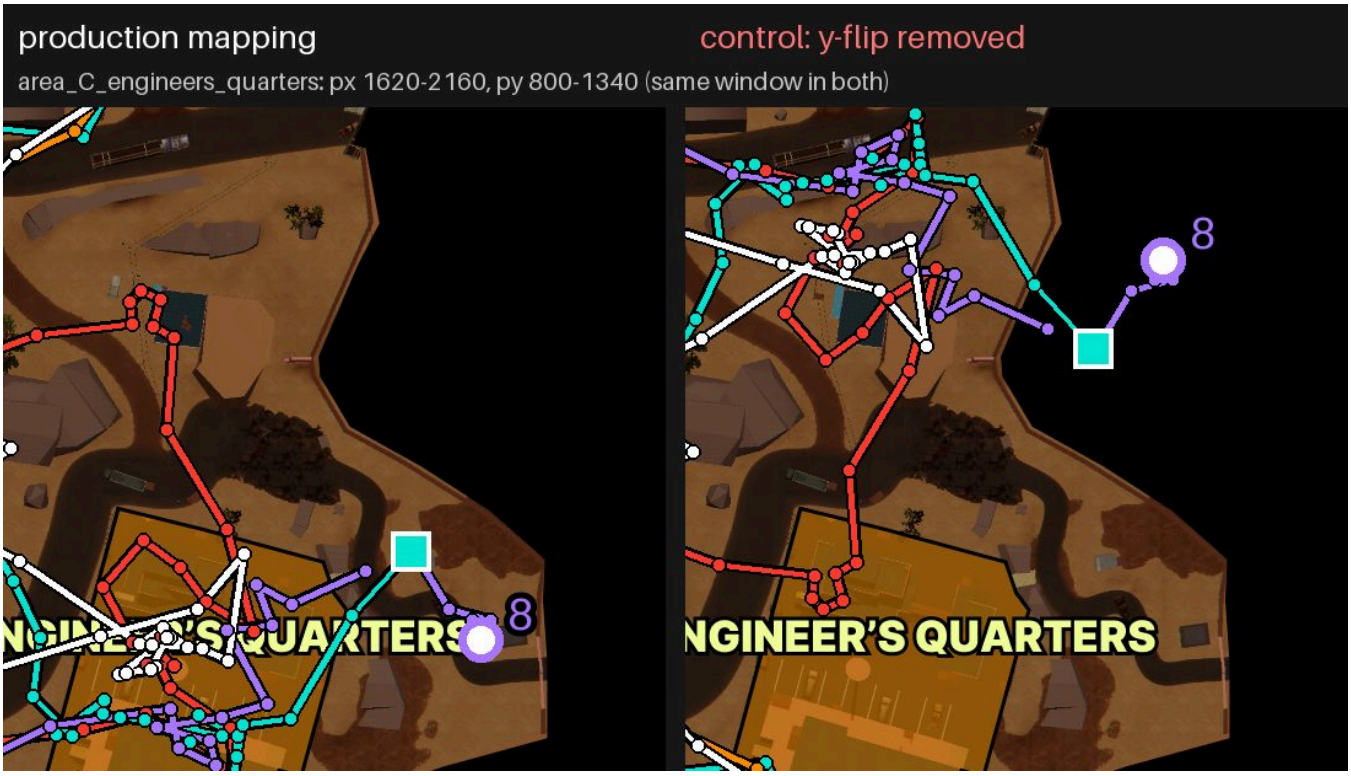
Burnt Zone crop. Journeys 2, 6, 7 and 10 start together on map ground; without the flip the area is empty.



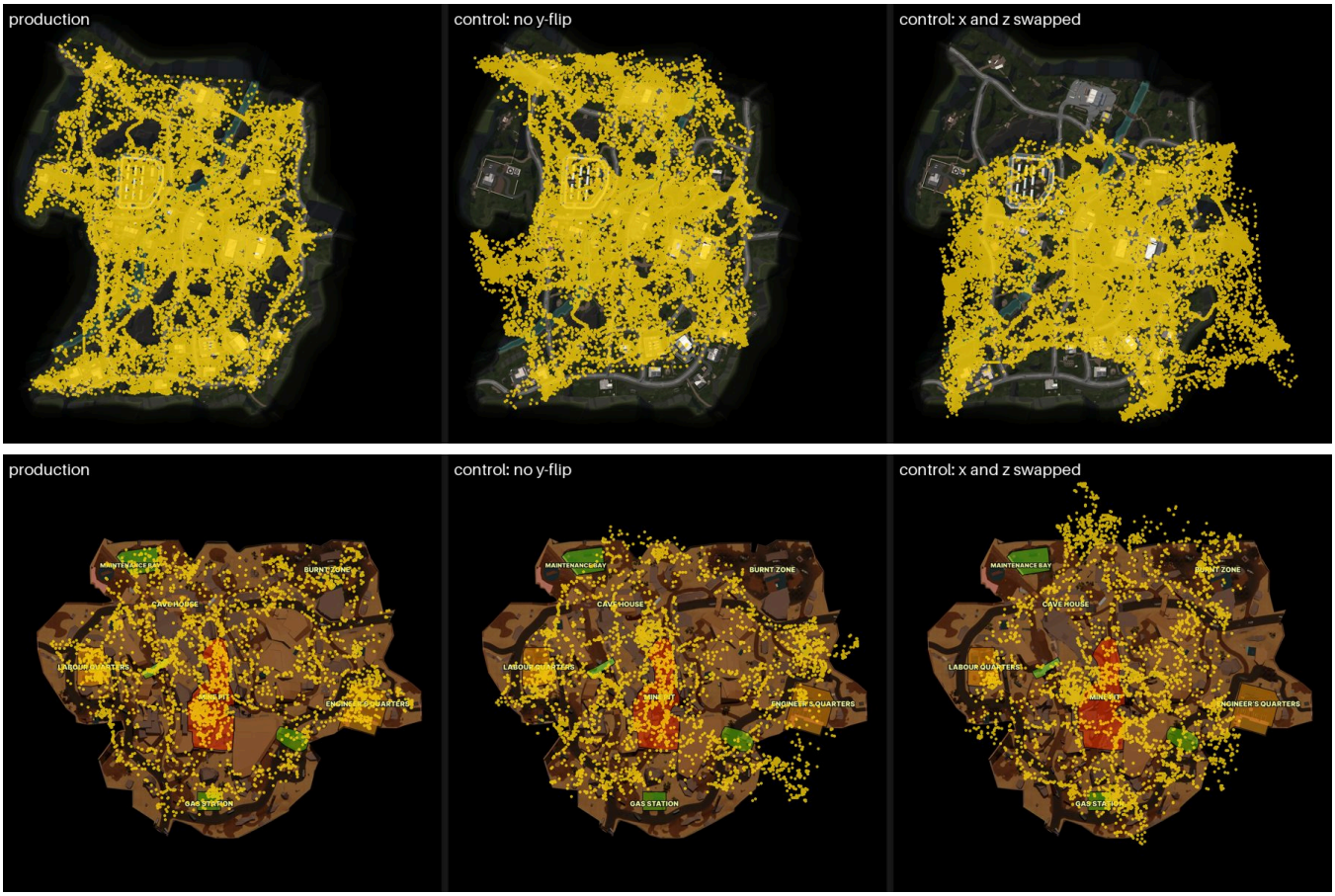
South-east crop. Without the flip, starts and dense paths sit in the black area outside the map.

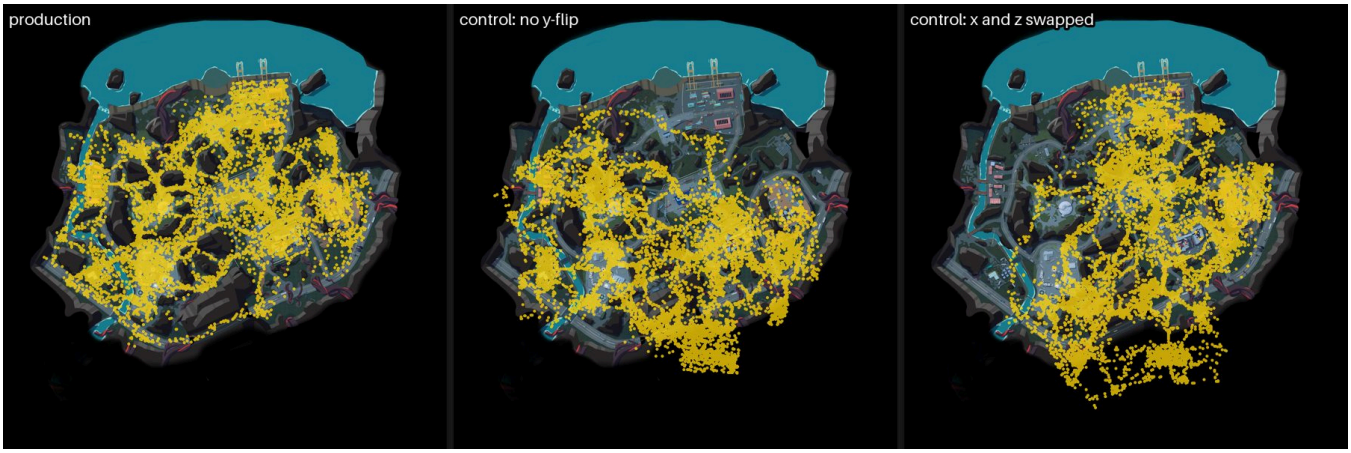


Engineer's Quarters crop. Journey 8's activity centres on the building; without the flip it shifts north into the void.



All three maps: production, no y-flip, and x/z swapped. Only production keeps positions on playable land.





Limits

- No landmark ground truth exists, so the mapping is *consistent with* the art rather than *verified*.
- Errors of a few pixels can't be seen.
- GrandRift's 2 px non-square difference (≤ 0.1 % y-stretch) is below what the method can detect.

Sources: grandrift_validation.md · grandrift_metrics.json · reverse_mapping_validation.md · RECONCILIATION_EVIDENCE.md §2

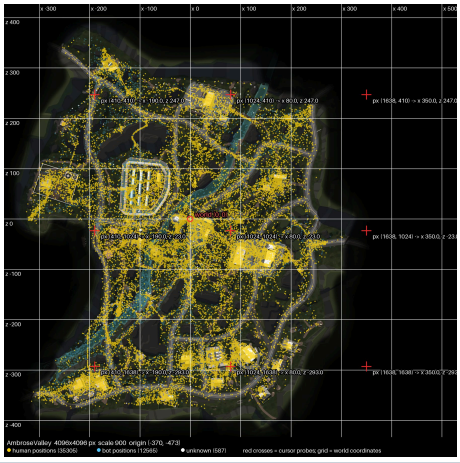
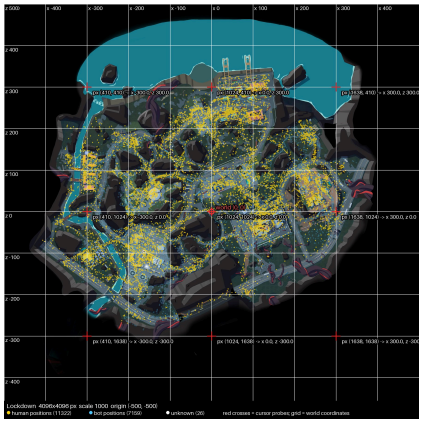
A4. The browser's reverse mapping (hover) is the exact inverse

Status: Current. **Assumption:** the pipeline stores UV. The browser converts UV back to world x/z for the hover readout, using each map's own width and height.

Evidence

- **Round trip:** all **72,676** exported path points return to their parquet x/z. The maximum error is the 4-decimal rounding limit and nothing more: 0.0636 / 0.0409 / 0.0707 world units on AmbroseValley / GrandRift / Lockdown, at most about 0.29 of a WebP pixel.
- **Pixel probes:** 2,005 per map. The differences are floating-point noise (about 1e-13).
- **Extent:** all 72,996 recorded positions fall inside their map's calibrated square. None is clamped.
- **Proportions:** GrandRift is the only non-square map. Treating it as square would put the south edge 2 px off. The code uses W and H separately.

Overlays: every exported position (humans yellow, bots blue; white = the 17 journeys then classed `unknown`), a 100-unit world grid and cursor probes labelled with the world x/z the browser returns.

AmbroseValley	GrandRift	Lockdown
		

Limit: deck.gl's `BitmapLayer` placement was not exercised in a running browser.

Sources: reverse_mapping_validation.md · reverse_mapping_metrics.json

A5. Duplicates: one file read once, three doubled journeys halved, nothing else removed

Status: Current (decision D3; EC-04, EC-29, EC-06).

Evidence

- **One byte-identical file** (same SHA-256) is in both `February_10` and `February_11`. The pipeline reads 1,242 of 1,243 files.
- **Three journeys hold their content twice.** In each, the second half equals the first half on all 8 columns, and `ts` jumps backwards at the midpoint. They have 8, 78 and 294 rows; **190 rows** are removed. Without this, 2 of them would show two deaths each. These are the only files whose rows are out of time order.
- **Identical same-second Loot/BotKill rows are kept (1,227 rows).** Same-second loot continues a smooth pickup-rate curve: 1,273 loot→loot gaps of 0 s, 1,590 of 1 s and 664 of 2 s. Loot rows 1–3 s apart are a median 0.10 units apart.
- **Open uncertainty:** if those rows were logging repeats, loot per journey would fall 7.6–14.5 % (most on GrandRift: 15.37 → 13.14). This can't be decided from the data.
- **Pipeline totals:** 89,016 rows read → 88,826 after un-doubling → 88,653 final.

Sources: build_report.json · INDEPENDENT_EVIDENCE.md §2.2–2.4 · RECONCILIATION_EVIDENCE.md §7.4

A6. Actors are classified by behaviour, not by ID

Status: Current (EC-09; changed 2026-10-24). **Assumption:** a file that logs `BotPosition` is a bot; every other file is human. 17 journeys (3 actors) have numeric "bot" IDs but log `Position`/`Loot`. They count as **human** and stay flagged `id_mismatch`. Totals: **798 human, 444 bot, 0 unknown**.

Evidence that the 17 come from the player client

Event	UUID files (781)	Numeric + `BotPosition` (444)	Numeric, no `BotPosition` (17)
`Position`	781	0	17
`BotPosition`	0	444	0
`Loot`	724	**0**	14
`KilledByStorm`	39	0	0

- No file logs both `Position` and `BotPosition`.
- **Speed:** no bot ever exceeds 5.0 units/s, across all 444 bot files. 12 of the 17 do, as do 728 of the 781 UUID journeys.
- **Role in the match:** all 17 start at their match's first event, as every UUID journey does. Only 28 of 444 bot journeys do.

Evidence against "a person played them"

- The ID format says bot.
- Actor A (14 of the 17 journeys) is an ordinary bot in 3 other matches.
- Only 1 of the 17 dies, against 442 of 781 UUID journeys. Their median length is 161 s, against 367 s.

Why "human" was still chosen (S, 2026-10-24)

- The events say player client, so they are counted.
- The cost: nothing proves a person played them.
- The effect is small. Adding them changed heatmap totals by at most **1.63 %** (AmbroseValley traffic). Every layer kept its top-10 cells, with per-cell correlation ≥ 0.9993.
- Human journeys went 781 → 798, players 245 → 248, and every match now has a human.

Map	Layer	Share from the 17	Top-10 cells shared	Pearson r
AmbroseValley	kills / deaths / traffic	0.78 % / 0.32 % / 1.63 %	10/10 each	≥ 0.9993
GrandRift	kills / deaths / traffic	0 % / 0 % / 0.62 %	10/10 each	≥ 0.9997
Lockdown	kills / deaths / traffic	0 % / 0 % / 0.24 %	10/10 each	≥ 0.9999

Sources: event_forensics/actor_classification.md (written when the class was `unknown`) · actor_classification.csv (regenerated: `final_classification = human`) · heatmap_impact.md · INDEPENDENT_EVIDENCE.md §3

A7. An event's position is the file owner's position

Status: Current (EC-11). **Assumption:** a kill marker shows where the killer stood, and a death marker shows where the victim fell. The other party's position is never in the row.

Evidence

- **1,686 of 2,198** Loot rows that share a second with a Position row have identical x/z (median distance 0).
- **Cross-file pairing:** a human `BotKill` and a bot `BotKilled` within ±2 s sit a median **12.6 units** apart. That is two different owners' positions, not one shared spot.
- **Side finding:** bot files use `BotKill`/`BotKilled` from the bot's own point of view, including **bot-on-bot kills (43 pairs)**. No published output depends on this.

Sources: INDEPENDENT_EVIDENCE.md §5.1–5.3

A8. `Kill`/`Killed` pairs: keep both, claim no cause

Status: Current (EC-12). **Assumption:** the 3 pairs are kept with their original labels and flagged `kill_killed_pair`. The `Killed` counts as a death; the `Kill` is left out of kill counts. They are never called self-kills.

Evidence

Case	Map	Match time	Same second and same x/y/z	Journey ends	Other recorded files in the match
P1	GrandRift	431 s	yes	1 s later	0
P2	AmbroseValley	546 s	yes	2 s later	0
P3	AmbroseValley	639 s	yes	at once	0

Why "self-kill" isn't supported

- Identical positions are expected whatever happened (see A7).
- 743 of 796 matches record only one file, so an unrecorded opponent can't be ruled out.
- No row has a victim or killer column.
- The pairs are logged together: no `Kill` occurs without a `Killed` in the same second, while only 2 of 400 `BotKilled` seconds also have a `BotKill`. That points to one incident logged twice. It says nothing about the cause.

Sources: kill_killed_pairs.md · kill_killed_pairs.csv · INDEPENDENT_EVIDENCE.md §5.4

A9. Positions after a death are dropped

Status: Current (D2, EC-14). **Assumption:** playback must not show a dead player moving. Position rows in a later second than the death are dropped. The 2 late `BotKill`s are kept and flagged `after_death`.

Evidence

- 173 positions are dropped across 172 journeys; 2 `BotKill`s are flagged.
- Post-death positions move a median **0.10 units** (max 1.65) from the death point, which fits a body or camera settling.
- No file has more than one death once doubling is undone (A5).
- 33 same-second positions after a death remain. The audit calls this negligible.

Sources: build_report.json · INDEPENDENT_EVIDENCE.md §5.7 · RECONCILIATION_EVIDENCE.md §3

A10. A match's date is the UTC date of its first event

Status: Current (D1). **Assumption:** this date is labelled the "observed start date". 2026-02-09 and 2026-02-14 are flagged as boundary and partial dates.

Evidence

- Folder date equals the UTC first-event date for 1,232 of 1,243 files. The 11 exceptions start 23:51–23:58 UTC and sit in the next day's folder.
- Matches per date: 02-09: 1 · 02-10: 287 · 02-11: 197 · 02-12: 162 · 02-13: 112 · 02-14: 37.
- Against the folder rule, **4 matches** change date. The audit calls this negligible.

Sources: build_report.json · INDEPENDENT_EVIDENCE.md §4.4 · FORENSIC_AUDIT_REPORT.md B5

A11. Gaps over 30 s break paths, and traffic is traced along the line

Status: Contested (D5, EC-16, EC-28; audit finding A1). **Assumption:** a gap between samples means where the player went is unknown. So paths break at gaps over 30 s, traffic spreads step time along the A→B line, and longer gaps are dropped.

What the audit found

- **Rows come on a fixed 5 s tick.** 99.92 % of human gaps over 6 s are within ±1 s of a multiple of 5.
- **No movement step has zero displacement** (0 of 71,607), so a missing row means the actor wasn't moving.
- **Displacement doesn't grow with gap length.** Human medians: 13.1 units over 5 s, 9.0 over 30–60 s, 6.9 over 2–10 min. Random row loss would make it grow.
- **Events inside gaps sit next to the earlier point A.** For gaps over 30 s: median 0.85 units from A; 94.3 % closer to A than to B; none further than |AB| + 5.

What this means

- A gap of g seconds is mostly (g − 5) s standing near A, not unknown movement.
- The production rule omits **3.9–4.3 %** of human time.
- Some standing-spot cells are under-counted by up to 7–16×.
- Across the plausible models, region magnitudes move ±1–16 %.

What still holds

- The overall traffic picture: Spearman 0.983–0.996.

- The **strongest traffic region on each map** is unchanged under all six models.

Fix direction (not applied; S decides)

- Put (g − 5) s at A, with no cut-off.
- Rename "estimated movement time".

Sources: FORENSIC_AUDIT_REPORT.md A1 · FOLLOWUP_EVIDENCE.md §1 · RECONCILIATION_EVIDENCE.md §7.1 · `d01_emission_rule.json`, `d02_allocation_models.json`, `c02_traffic_sensitivity.json`

A12. Bot journeys are recorded in only some matches

Status: Current wording (audit finding C1, corrected in the Decisions table). **Assumption:** say "bot journeys recorded in 6.5 % of matches", never "bots in 6.5 % of matches".

Matches (of 796)	Count
With recorded bot files	**52** (6.5 %)
With a human-file event labelled as combat with a bot	**722** (90.7 %)
… where a human was killed by a bot	399
With no evidence either way	54

Do the labels really point to bots? In the 52 matches with bot files, a human's bot-combat event pairs with a bot file's matching event in **25 %** of cases. By chance it would be 1.6–3.2 %. Every pairing is with a bot file, and none with another human.

Sources: FOLLOWUP_EVIDENCE.md §4 · `d04_bot_presence.json`

A13. Rates are per recorded human journey

Status: Current (stats wording). **Assumption:** most matches record one player, so rates use "human journeys" as the denominator, not matches.

Evidence

- **743 of 796** matches contain exactly one file.
- The README's "10 humans and 40 bots produce 50 files" is **contradicted** for this dataset.
- The frontend check found one label ("kills per match") used for two denominators at two precisions. It was fixed to "kills per human journey" (2.99, matching the stats pages) and "% of human journeys ending in a recorded death" (56.9 %).

Sources: RECONCILIATION_EVIDENCE.md §1 · frontend_verification/README.md

A14. Storm deaths stay in the Deaths layer

Status: Current (feature 6b). **Assumption:** storm deaths stay in the Deaths layer and their count is shown separately. There is no separate storm layer.

Evidence

- 39 storm deaths, all in human files.
- They happen **655–887 s** (median 739 s) into a match: none before minute 10, then 1 / 11 / 17 / 7 / 3 in minutes 10–14.

- They make up 17 of 315 deaths on AmbroseValley, 5 of 30 on GrandRift and 17 of 97 on Lockdown (**17–18 %** of the layer on the two smaller maps).
- Bot-caused deaths peak in minutes 1–3.
- **Note:** CLAUDE.md says "11–15 min", but one storm death is at **10 min 55 s** (audit C4).

Sources: INDEPENDENT_EVIDENCE.md §5.5 · spatial_analysis/REPORT.md §5

A15. Ending alive is not proof of extraction

Status: Current. **Assumption:** no extraction event exists, so anything built on "survived" is labelled *inferred*.

Evidence

- 339 human journeys end alive.
- Their spans are **86–840 s** (median 512 s), against a median of 286 s for journeys ending in death.
- An 86 s journey argues against reading every one as an extraction.

Sources: INDEPENDENT_EVIDENCE.md §5.6 · FORENSIC_AUDIT_REPORT.md B4

A16. Single 64 × 64 cells aren't stable; 16 × 16 regions are used for hotspots

Status: Current. **Assumption:** the 64 × 64 grid is kept as the numbers and the picture. Hotspots are named only at 16 × 16 regions (4 × 4 cells).

Evidence

- **Most kill and death cells hold one event.** For deaths, that is 72 % / 93 % / 82 % of occupied cells on AmbroseValley / GrandRift / Lockdown.
- **Split-half test:** the matches are split into two halves, and the test checks whether both halves pick the same strongest cells. For AmbroseValley kills, the top-5 overlap is **0.0** at 64 × 64, **0.6** at 32 × 32 and **0.8** at 16 × 16 (chance 0.0).
- **GrandRift deaths** show no repeatable pattern at any resolution: 0.0 / 0.0 / 0.2, against a chance level of 0.2.
- **External guidance:** a GameAnalytics post asks for "a few hundred samples" per bin. Our strongest 16 × 16 kill zones hold 139 / 19 / 36 events. Kill-vs-death balance maps are not attempted because the data records only one side of each event.

Sources: spatial_analysis/REPORT.md §0–2 · spatial_analysis.json · external_references.md

A17. Hotspot rule: ≥ 5 matches, no match over 50 %, at most 6 per layer

Status: Current since 2026-10-24. It **supersedes** the earlier "consistent concentration" rule: top decile, ≥ 0.8 stability over 300 resamples, ≥ 3 players, and surviving three half-region grid shifts.

Why the earlier rule was dropped

- **It was fragile.** The audit found the only AmbroseValley death hotspot at stability **0.779 [0.767–0.790]**, below its own 0.8 threshold. It was "consistent" in only 39 % of 100 seed runs.
- **The shift check behaved oddly.** It rejected two more stable death regions (0.896 and 0.845) and was the single most consequential rule. Without it, GrandRift would have shown 8 traffic regions instead of 3.
- **The thresholds mattered a lot.** With the top 5 % instead of the top 10 %, almost nothing was left; with the top 15 %, many more passed.

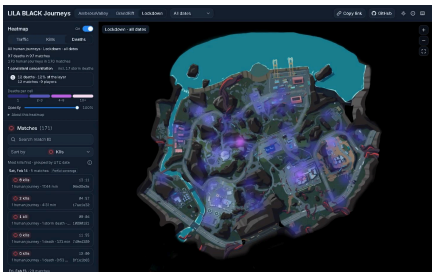
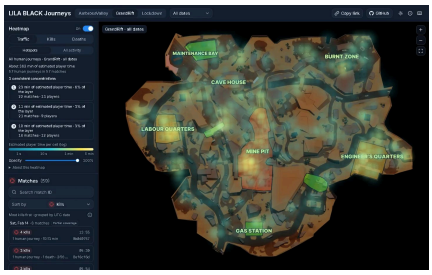
Before and after (57 selections)

Layer	Selections with no hotspot	Hotspots in total	Median share of the layer held by hotspots
Traffic	6 → 4	39 → 89	10 % → 24 %
Kills	16 → 9	7 → 53	9 % → 35 %
Deaths	17 → 13	2 → 29	9 % → 40 %

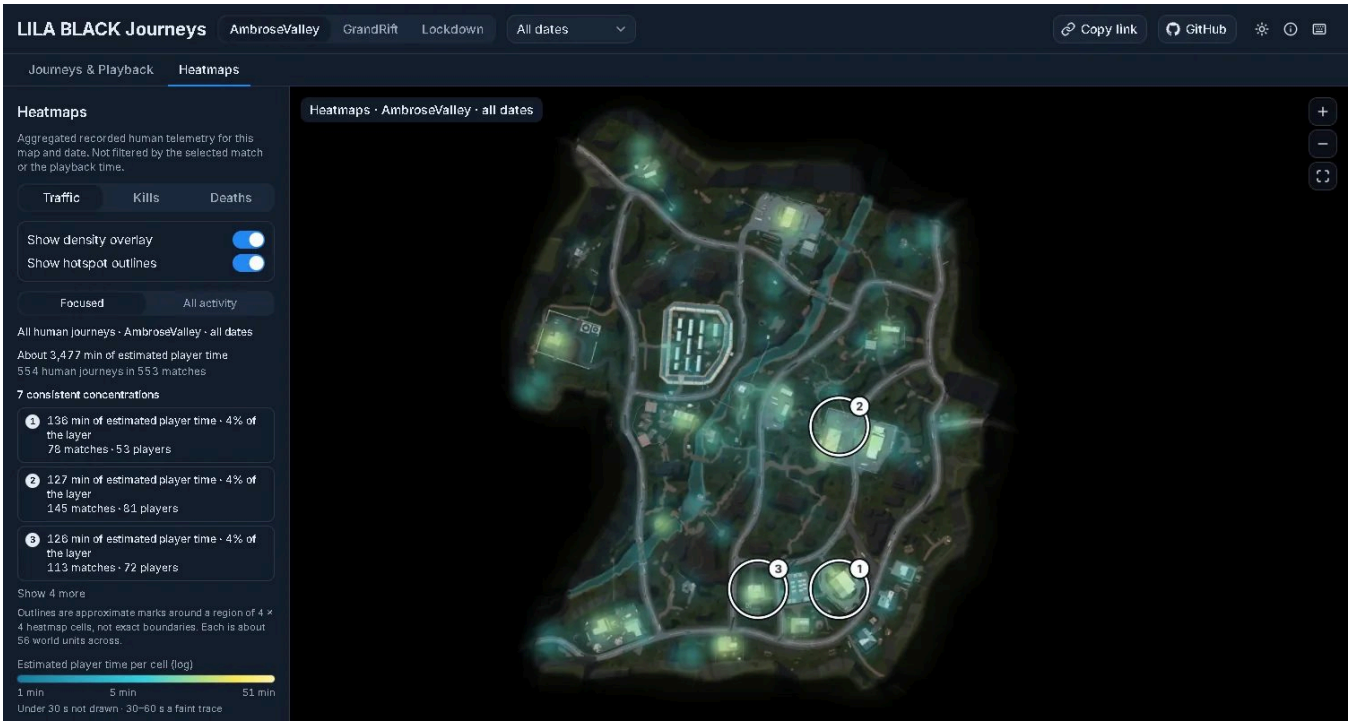
- Every earlier kill and death hotspot is kept, and 34 of 39 traffic hotspots.
- GrandRift deaths still has none: no region reaches 5 matches.
- **What is given up:** stability is no longer measured, so a hotspot can partly reflect where the region grid falls. The ≥ 5 matches and ≤ 50 % rules limit this but don't remove it.

Earlier rule, in the UI at the time: AmbroseValley kills, 5 consistent regions, region 2 selected.



GrandRift kills: none consistent	Lockdown deaths: 1 consistent	GrandRift traffic: 3 consistent
		

Hotspot outlines (feature 6c): a circle around each region, numbered strongest first.



Sources: hotspot_rules_compare.md · zone_sensitivity.md · spatial_analysis/REPORT.md §3, §8 · FOLLOWUP_EVIDENCE.md §2

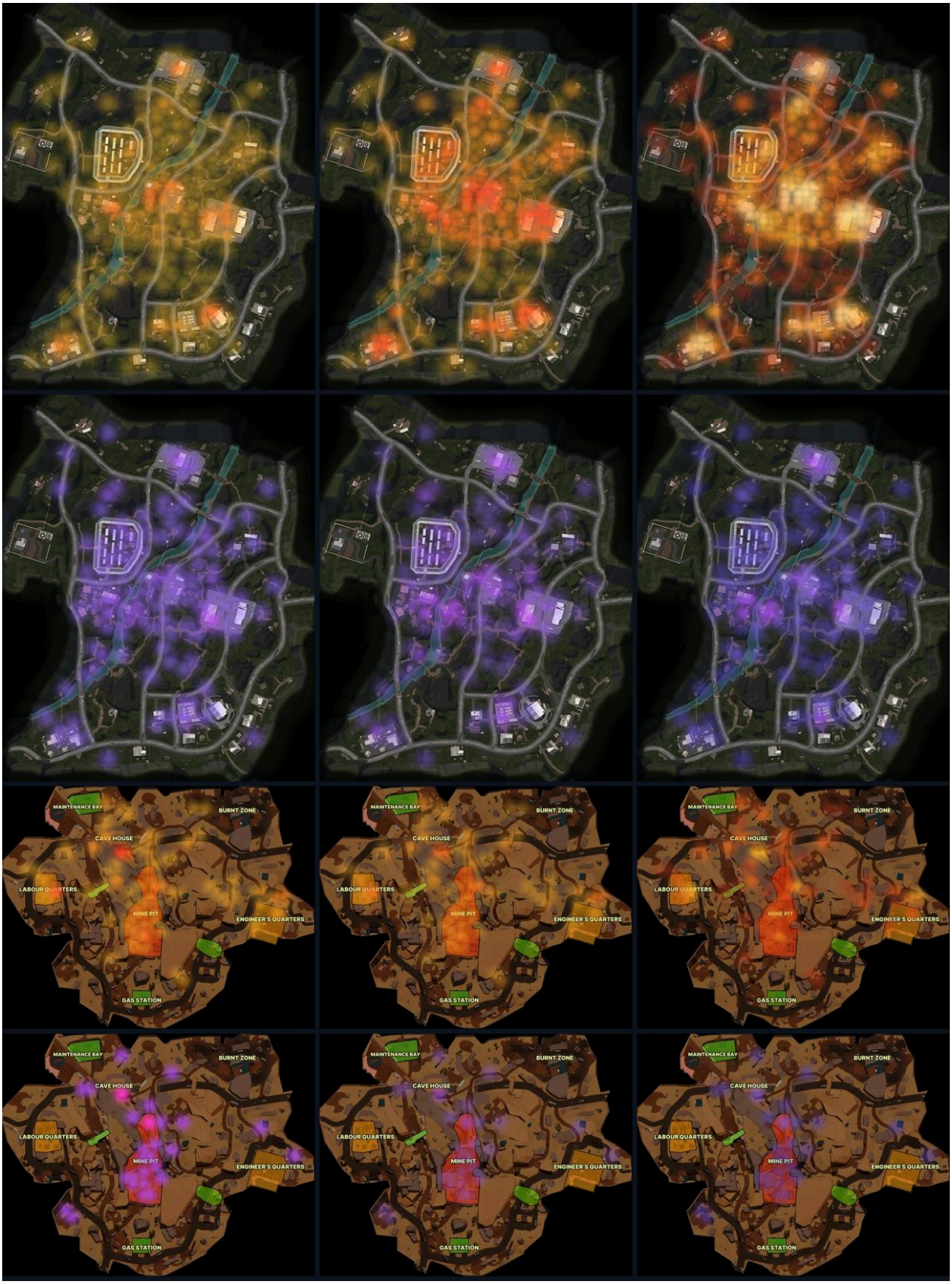
A18. Kills and deaths are coloured by count classes

Status: Current (feature 6b). **Assumption:** 64 × 64 kill and death cells are coloured by absolute count, **1 / 2–3 / 4–9 / 10+**, the same on every map and date. Lightness and opacity rise with the count.

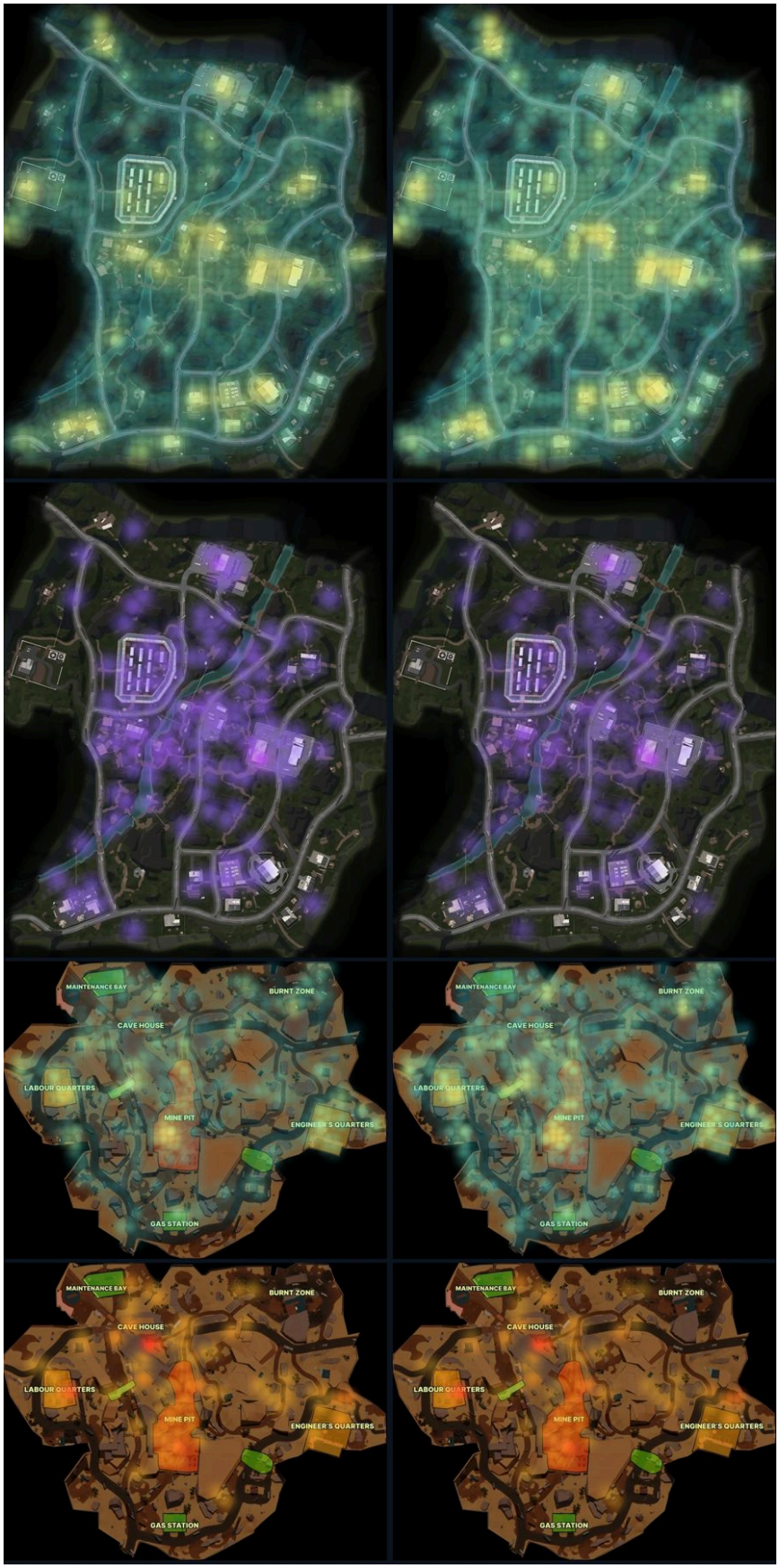
Evidence

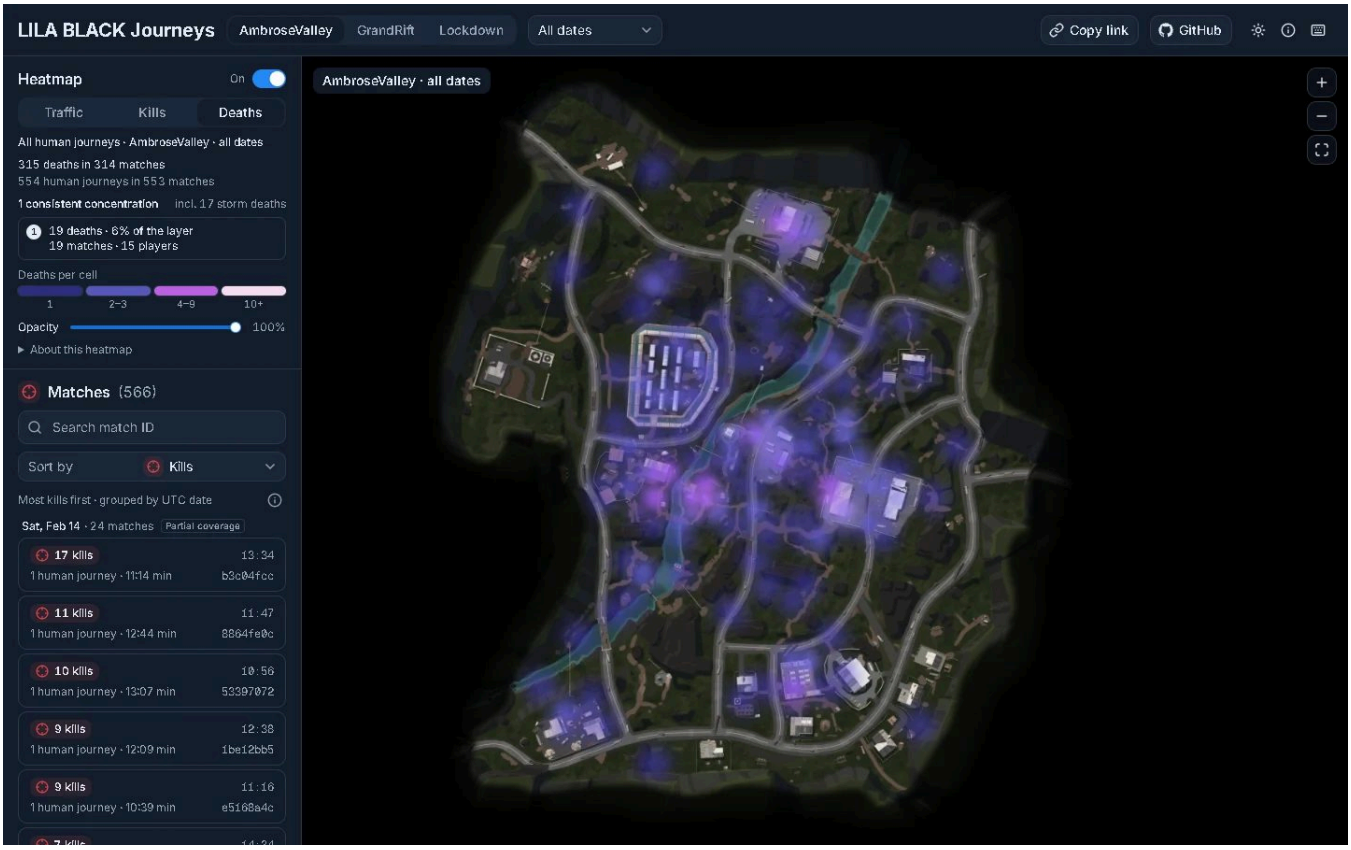
- **The continuous look didn't separate counts.** Neighbouring counts 1, 2, 4 and 10 were only 4–9 dE apart for kills, and 0 dE for GrandRift deaths (its maximum cell is 2).
- **Hue-only bands fail under colour-vision deficiency.** The same palette cut into bands collapses to 0.9–3.8 dE for kills.
- **Lightness-ordered classes hold up.** They stay ≥ **18.9 dE** apart for kills and ≥ **7.0 dE** for deaths under deuteranopia and protanopia, on every map's typical, darkest and brightest colours.
- **Fixed fractions of the maximum don't work for sparse counts.** 393 of AmbroseValley's 460 kill cells would fall in the bottom band.

Left: the previous continuous look; middle: the old palette cut into bands; right: the lightness-ordered classes (rows: AmbroseValley kills and deaths, GrandRift kills and deaths).



Continuous glow (left) vs four bands (right).


MAINTENANCE BAY
BURNT ZONE
CAVE HOUSE
LABOUR QUARTERS
MINE PIT
ENGINEER'S QUARTERS
GAS STATION
MAINTENANCE BAY
BURNT ZONE
CAVE HOUSE
LABOUR QUARTERS
MINE PIT
ENGINEER'S QUARTERS
GAS STATION
MAINTENANCE BAY
BURNT ZONE
CAVE HOUSE
LABOUR QUARTERS
MINE PIT
ENGINEER'S QUARTERS
GAS STATION
MAINTENANCE BAY
BURNT ZONE
CAVE HOUSE
LABOUR QUARTERS
MINE PIT
ENGINEER'S QUARTERS
GAS STATION



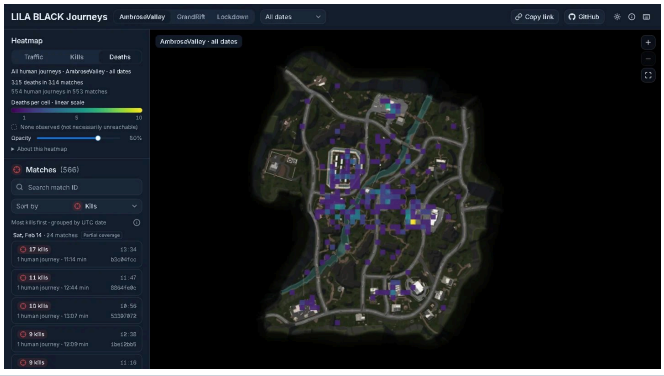
Sources: spatial_analysis/REPORT.md §4, §7 · count_bands.json · visual_encoding.json · external_references.md §3 (Machado et al. 2009)

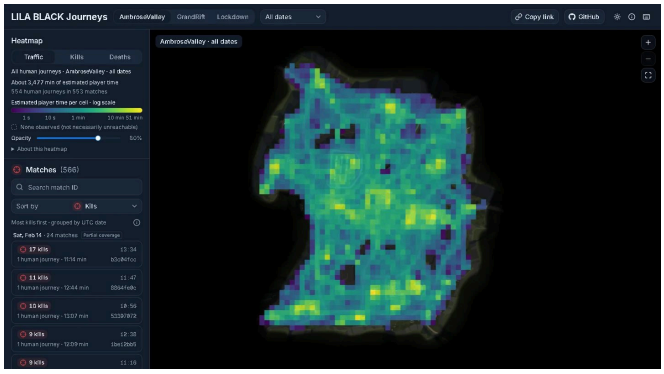
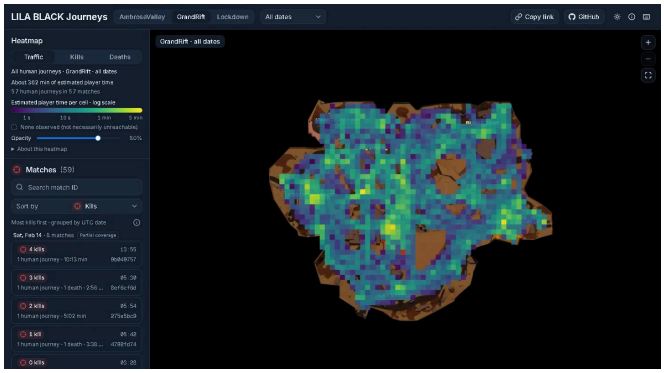
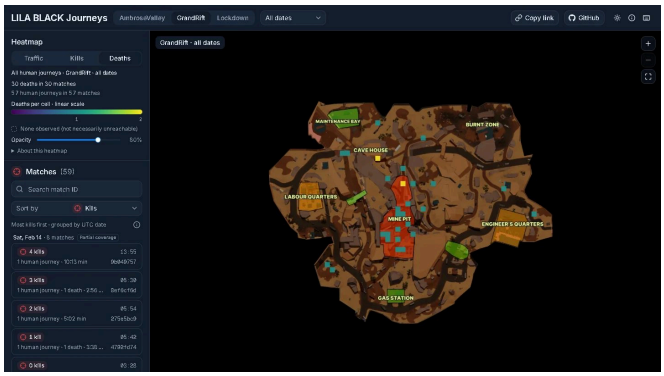
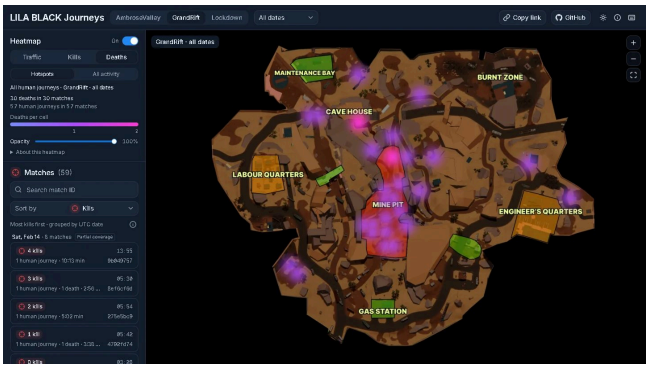
A19. Heatmaps are a painted picture, not a cell mosaic

Status: Current (feature 6, visual pass). **Assumption:** each cell is drawn as a soft bump (σ = 1 cell). Opacity follows intensity, and overlapping bumps keep the stronger value without adding up, so no peak is invented. The numbers and the hover stay exact.

Evidence (before and after screenshots)

- **Why it read as a mosaic:** traffic covers 40 % of AmbroseValley's cells, and every cell had the same opacity.
- **Deaths looked uniform:** 72 % of AmbroseValley's death cells hold one event, so they were all the same dark square.

Before	After
	

Before	After
	
	
	

Further states are in `heatmap_review/`: GrandRift traffic in All activity, the heatmap switched off, an empty date, a zoomed view, Lockdown deaths on one date, and a match open with the heatmap.

Note on one date: an absolute 30 s / 60 s traffic threshold (Focused view) hides **84 %** of GrandRift's cells (31 % of AmbroseValley's). So a single GrandRift day is nearly empty in Focused view, and "All activity" shows everything. This was left open.

Source: `docs/06_HEATMAPS.md` (visual pass) · `heatmap_review/`

A20. The map performance score

Status: Contested (audit finding B1). **Assumption:** five parts (journeys, median span, kill rate, loot rate, storm-death rate), each min–max scaled and weighted equally, give a relative ranking of the maps.

Variant	AmbroseValley / GrandRift / Lockdown	Same order in 1,000 bootstraps	AmbroseValley first
Published	0.76 / 0.49 / 0.25	82.2 %	97.3 %
Rank-based	1.4 / 1.0 / 0.6	82.7 %	97.7 %
Without human journeys	0.71 / 0.61 / 0.25	56.5 %	**66.4 %**

What the audit found

- **The order doesn't depend on how parts are scaled.** It does depend on which parts are included.
- **"Human journeys" inflates the stability.** It measures how much a map was played and has no sampling variance.
- **GrandRift overlaps both other maps on every part** at 95 % intervals. For example, its storm deaths per 100 are 8.77 [1.8–17.5].

Recommendation (S decides): drop "Leading" / "Needs improvement", or keep the score only as a labelled relative index.

Sources: FOLLOWUP_EVIDENCE.md §3 · `d05_map_score.json` · RECONCILIATION_EVIDENCE.md §7.3

A21. The site shows exactly the numbers the pipeline produced

Status: Current (verified 2026-10-24).

Evidence

- **Data files agree with each other: 8,011 checks, 0 mismatches**. This covers index ↔ stats ↔ heatmaps ↔ zones ↔ match files.
- **Rendered pages match: 815 checks, 0 mismatches**, 0 page exceptions. Headless Chrome was run on every map × date, with the expected text computed independently from the JSON.
- **One label defect was found and fixed:** "kills per match" (described in A13).

Sources: frontend_verification/README.md · `sources_report.json`, `rendered_report.json`

A22. The pipeline is mathematically correct and reproducible

Status: Current.

Evidence

- **Every published number was reproduced independently.** All 796 index rows, every match file and every kill and death grid match cell by cell. All **7,792 of 7,792** values in `stats.json` match, and a negative control caught the planted errors.
- **A rebuild is byte-identical,** WebP files included.
- **All 125 tests pass.** Most data expectations in them are counts taken from the same data. They catch regressions, not wrong assumptions; the audit's independent counts are the external check.

Sources: FORENSIC_AUDIT_REPORT.md · RECONCILIATION_EVIDENCE.md §4–6 · MANIFEST.md

Appendix A. Open points from the audit

Status as recorded in the audit, compared with today's CLAUDE.md.

ID	Point	Status now
A1	Traffic mis-places standing time and omits 3.9–4.3 % of it	Open (A11)
A2	AmbroseValley death hotspot failed its own stability rule	Moot: the stability rule was replaced (A17)
B1	Map score isn't a quality measure	Open (A20)
B2	Identical same-second loot rows: loot rate uncertain by 7.6–14.5 %	Open (A5)
C1	"Bots in 6.5 % of matches" wording	Fixed in the Decisions table (A12)
C2	Spec 02 point 15 says "355, shortest 25 s"; should be 339 and 86 s	Check spec 02
C3	EC-20 "alpha 255 on every pixel": 1,194 AmbroseValley pixels are below 255	Check EC-20 (immaterial)
C4	"Storm deaths 11–15 min" (one at 10:55); "gaps up to 8 min" (human max 5 min; 8.6 min is a bot gap)	Still in the CLAUDE.md Data facts
C6	`map_score` rounds before scaling (≤ 0.0003)	Open, negligible
C8	`INSIGHTS.md` stale	Check
C9	README says `make data` takes ~45 s; it took 127 s	Check README

Appendix B. Index of every evidence file

Folder / file	What it is
`build_report.json`	Pipeline counts: files, rows, duplicates, doubled journeys, actors, dates
`grandrift_validation.md`, `grandrift_metrics.json`, `grandrift_*.png`	Forward-mapping check on GrandRift with a no-flip control (A3)
`reverse_mapping_validation.md`, `reverse_mapping_metrics.json`, `reverse_*.png`	Reverse (hover) mapping on all three maps (A3, A4)
`event_forensics/`	Actor classes (A6), Kill/Killed pairs (A8), heatmap impact
`spatial_analysis/`	Sparsity, stability, resolution, colour encoding, count bands, external references (A16–A18)
`hotspot_rules_compare.md`	Old vs new hotspot rule (A17)
`heatmap_review/`	Before and after screenshots of the heatmap visual pass (A19)
`interpretation_layer/`	Screenshots of the earlier hotspot layer (A17, A18)
`tabs_and_outlines/`	Screenshots of the tabs, density and outline switches, and click-to-zoom (13 images)
`forensic_audit/`	Independent audit: Part 1 independent, Part 2 reconciliation, Part 3 follow-up, plus scripts (A1, A11, A12, A20, A22)
`frontend_verification/`	Data files ↔ rendered pages check (A21)